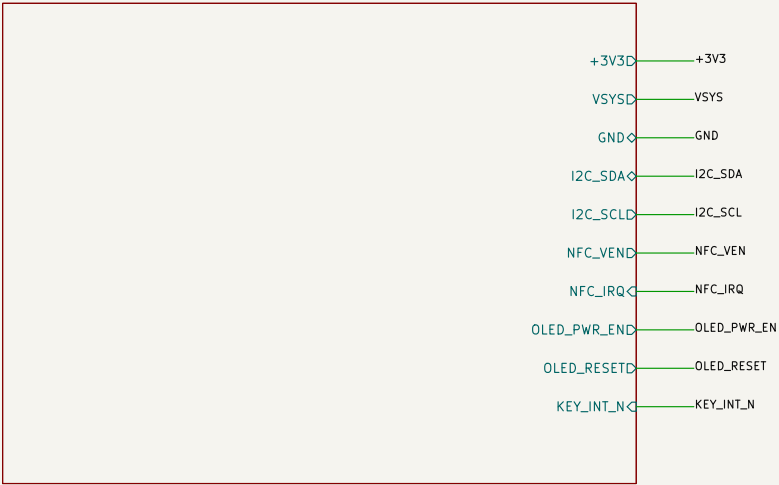
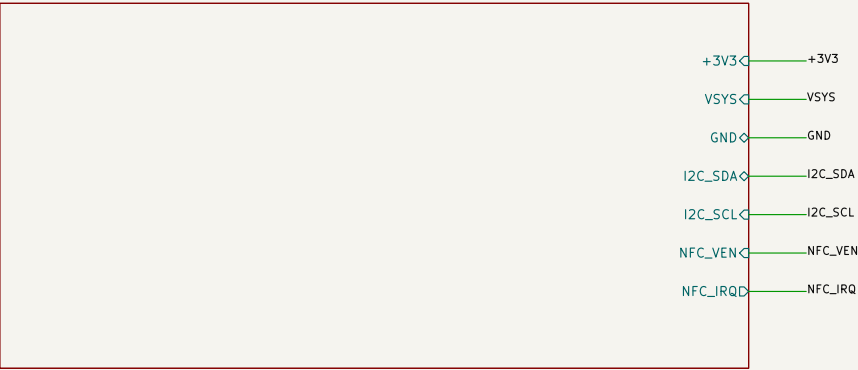


NUCULA v2 / system interconnect

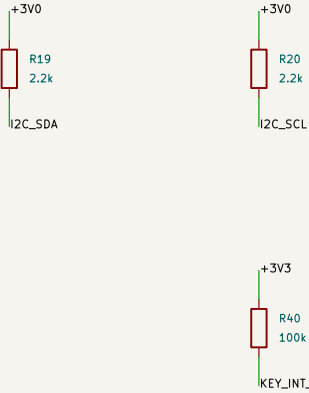
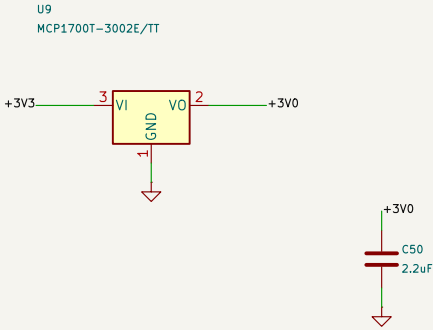
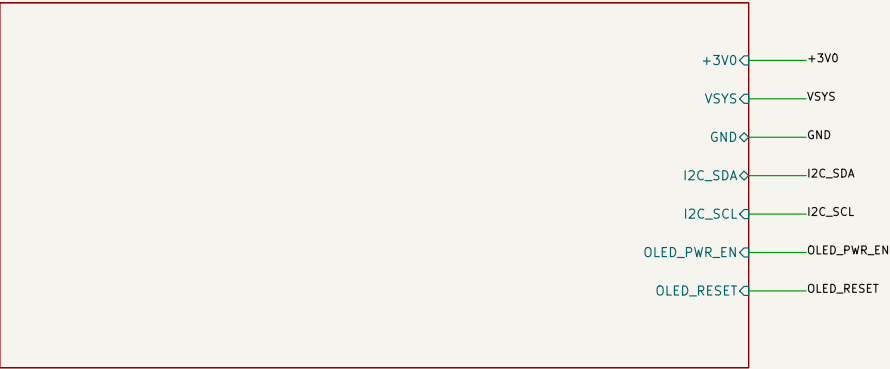
Power + ESP32-C3



NFC / PN7160 / 0x28

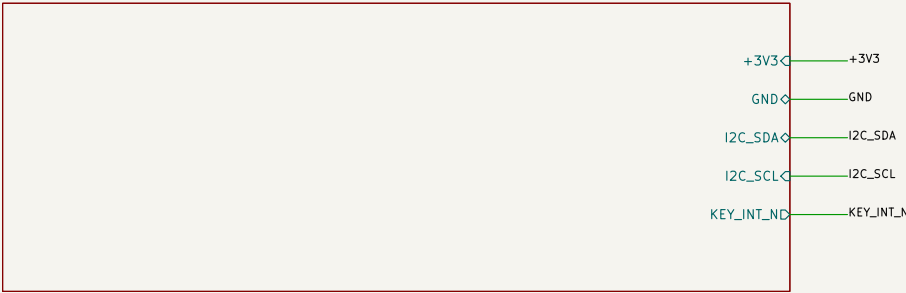


OLED / SSD1309 / 0x3C

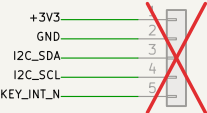


3.0 V display logic and common I2C pull-ups.
SSD1309 logic max = 3.3 V; existing buck worst case ~3.39 V.
3.0 V HIGH levels also meet the 3.3 V devices' thresholds.
One populated pair of I2C pull-ups. Bus speed: 100 kHz.
No I2C multiplexer is needed; addresses are distinct.

Keyboard / breakaway / 0x20



J4
MAIN LINK / OPTIONAL



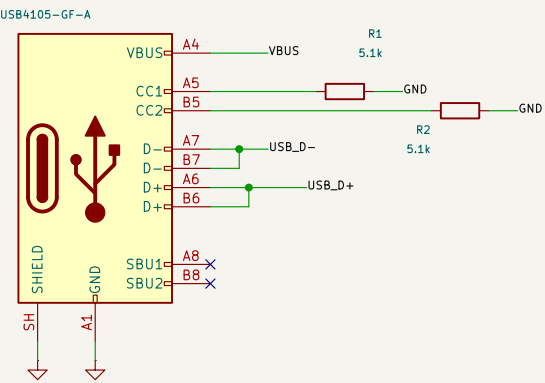
Break line carries five nets only.
Reconnect J4 <-> J5 after separation.
J3 9-pin keyboard header remains DNP.
Mechanical break line is deferred to PCB layout.

NFC / OLED power comes from VSYS; digital logic from regulated rails.
Concurrent Wi-Fi + NFC + OLED can exceed a small 400 mAh pack's rating.
Confirm peak battery current and USB input power before PCB release.
RF matching/coil and exact OLED flex remain intentionally provisional.

Sheet: / File: nucula-v2.kicad_sch		
Title: Nucula v2 - system interconnect		
Size: A3	Date: 2026-09-20	Rev: B-draft
KiCad E.D.A. 10.0.6		Id: 1/5

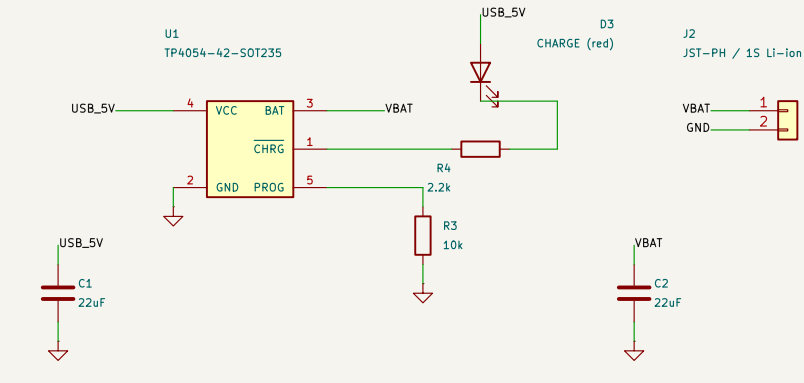
NUCULA v2 / USB-C, Li-ion power and ESP32-C3

01 USB 2.0 device / 5 V sink



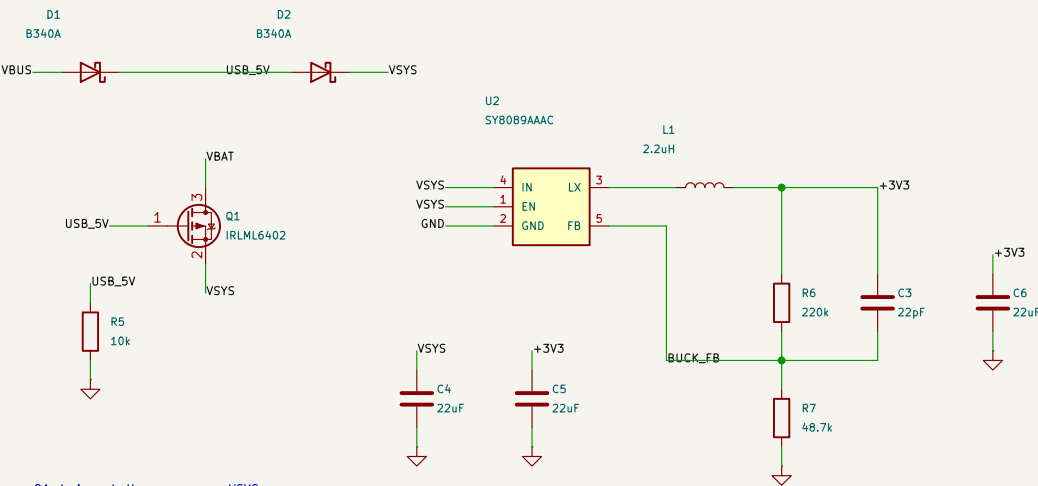
CC1 / CC2: independent 5.1k Rd. No USB-PD.
Route D+/D- as a 90 ohm differential pair.
Place U4 at connector; R12/R13 at module.
See design notes for USB current / inrush limits.

02 4.2 V / 100 mA charger



Protected 1S LiPo, ~400mAh (3.7V nom / 4.2V full).
J2: JST-PH 2.0mm; pin 1 = BAT+; pin 2 = GND.
100mA charge = 0.25C at 400mAh; ~10mA termination.
Confirm pack polarity and ratings; no cell-temp input.

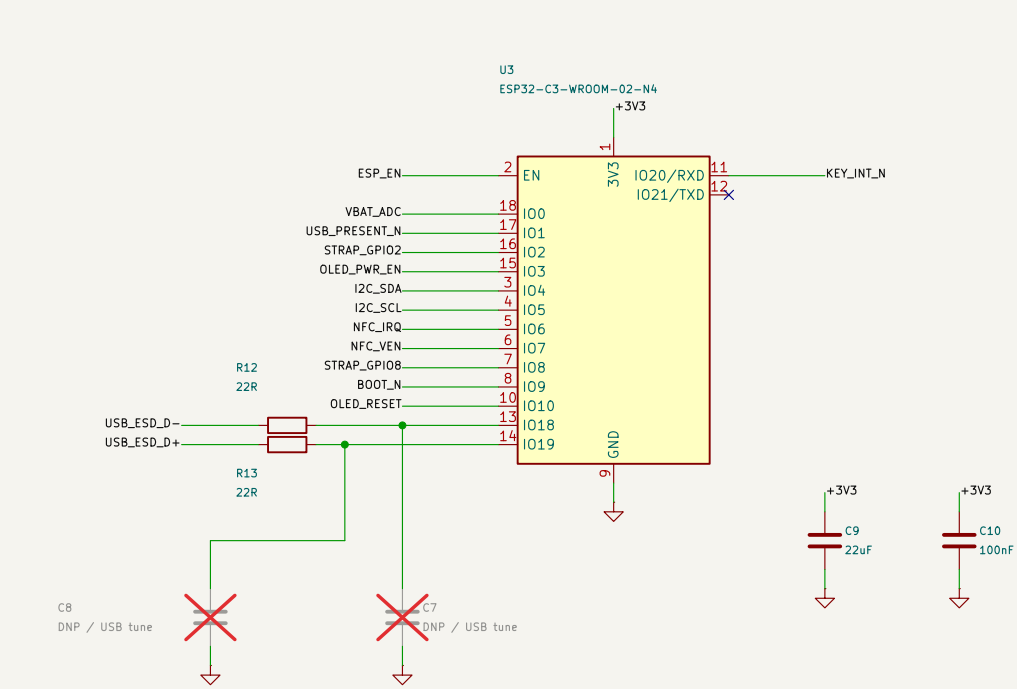
03 USB-priority supply / 3.31 V buck



Q1 drain = battery, source = VSYS.
Body diode points VBAT -> VSYS.
USB load bypasses charge-current sensing.

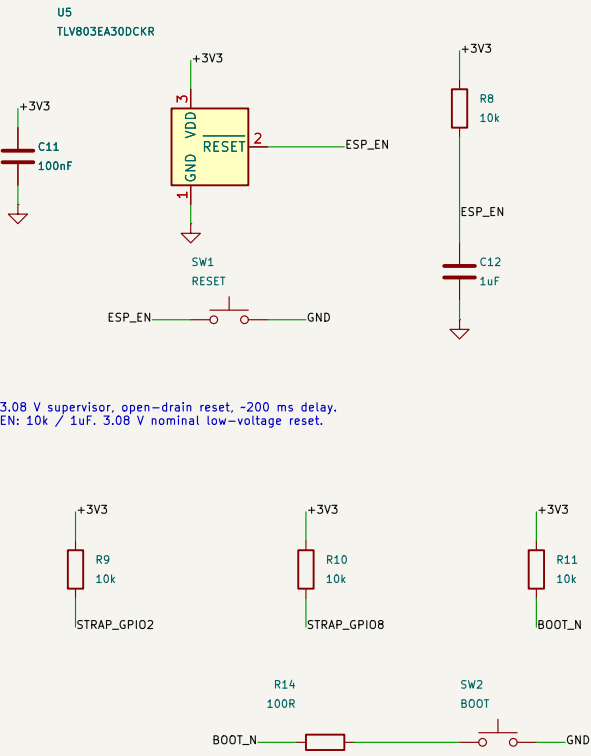
VOUT = 3.31 V; R6 / R7: 0.1% precision.
500 mA rail design targets validate transients.
Buck only: low battery causes dropout / reset.
C5 + C6: 44 uF nominal; verify DC-bias derating.

04 ESP32-C3-WROOM-02-N4 / native USB



USB D- = GPIO18; D+ = GPIO19. Native USB.
I2C: GPIO4 SDA / GPIO5 SCL; use 100 kHz.
GPIO6 NFC_IRQ; GPIO7 VEN; GPIO3 OLED power.
GPIO10 OLED reset; GPIO20 keyboard interrupt.
GPIO21 is reserved; straps GPIO2/8/9 unchanged.

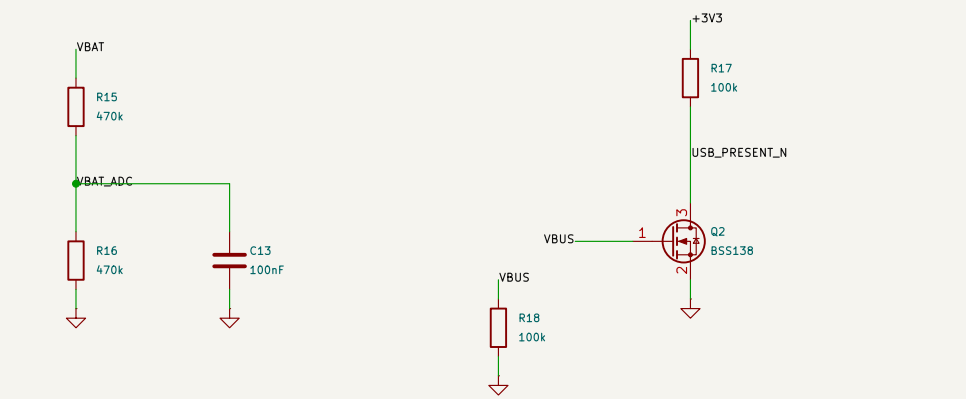
05 Reset, undervoltage and boot straps



3.08 V supervisor, open-drain reset, ~200 ms delay.
EN: 10k / 1uF, 3.08 V nominal low-voltage reset.

Hold BOOT; tap RESET to enter ROM download mode.
GPIO8 must stay high while GPIO9 is low.
No large capacitor on GPIO9.

06 Battery and USB presence sensing



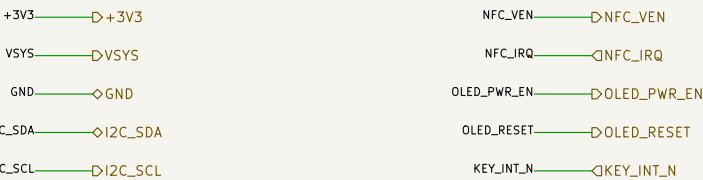
VBAT_ADC = VBAT / 2, max ~2.15 V incl. tolerances.
ADC1 GPIO0: use calibrated 11/12 dB attenuation.
Divider load ~4.5 uA; allow >=120 ms to settle.

USB_PRESENT_N: GPIO1 LOW when VBUS is present.
Firmware must disable USB attach when VBUS is absent.
Q2 isolates 5 V VBUS from the 3.3 V GPIO supply.

DRAFT / before PCB:
Confirm protected battery and JST polarity.
Validate USB current, inrush, suspend and ESD.
Verify regulator layout, capacitor bias and load steps.
Reference: Olimex ESP32-C3-DevKit-Lipo Rev C.
See docs/power-design.md for sources and limits.

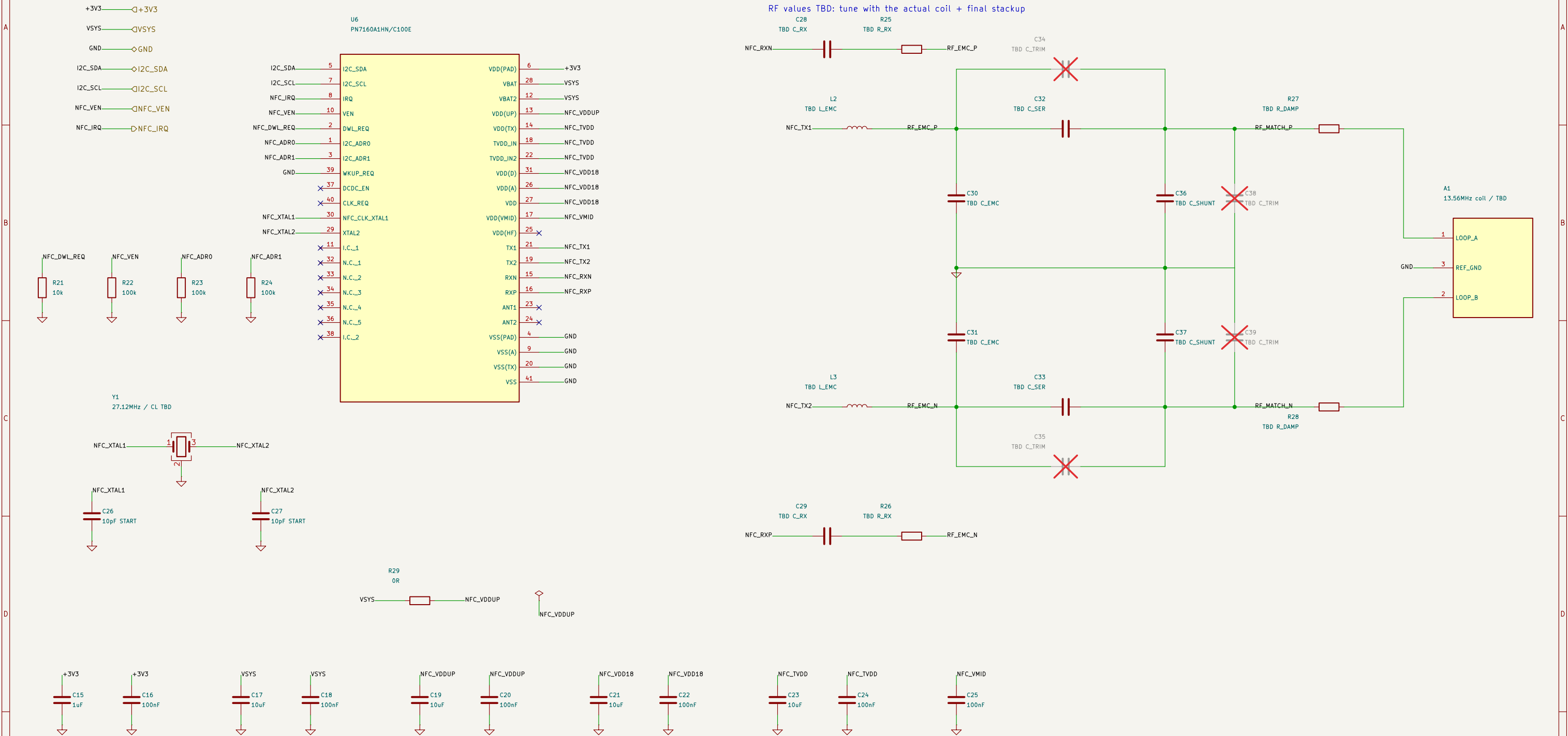


ERC source declarations



Sheet: /Power + ESP32-C3/ File: power-mcu.kicad_sch		
Title: Nucula v2 - USB-C, battery power and ESP32-C3		
Size: A3	Date: 2026-09-20	Rev: B-draft
KiCad E.D.A. 10.0.6		Id: 2/5

PN7160 / I2C host + balanced NFC antenna matching

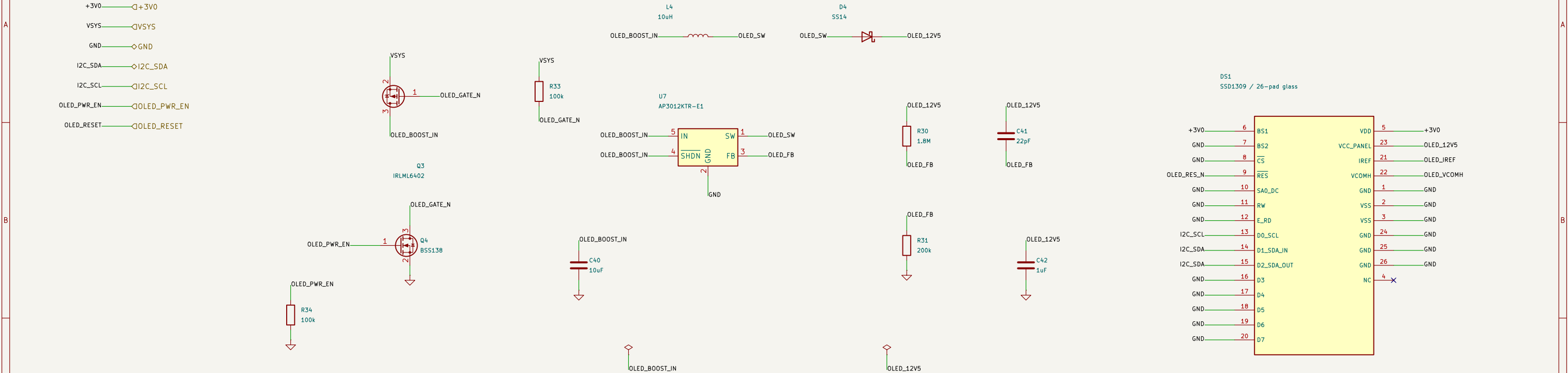


Pin 14 -> 18 + 22: internal TX-LDO output, not an external supply.
Pin 31 -> 26 + 27: internal 1.8 V rail; decouple only.
VSYS powers VBAT/VBAT2 and VDD_UP; VDD_PAD uses +3V3.
VEN defaults low. DWL_REQ test access via its labelled net.
ADR1:0 = 00 => 0x28. WKUP_REQ low: wake over I2C.
VEN defaults low. DWL_REQ test access via its labelled net.
ANT1/ANT2/VDHF unused; configure normal RF detection.
27.12 MHz crystal; confirm ESR, drive and load before ordering.

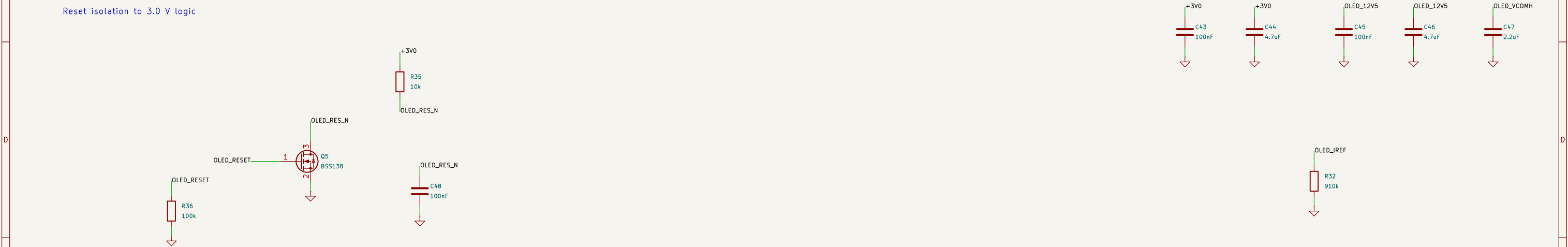
A1 pads 1/2 = coil ends; pad 3 = reference/shield ground.
No center-tap assumption. Coil footprint is a placeholder.
C34/C35/C38/C39 are optional parallel tuning pads (DNP).
All other RF values are unselected; do not fabricate yet.
Source: parts documentation/PN7160_schematic-3.pdf

2.42-inch OLED glass / SSD1309 / I2C

Switched OLED supply / default OFF



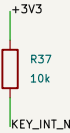
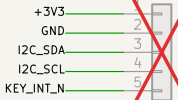
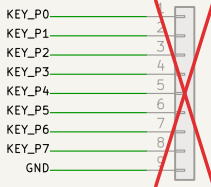
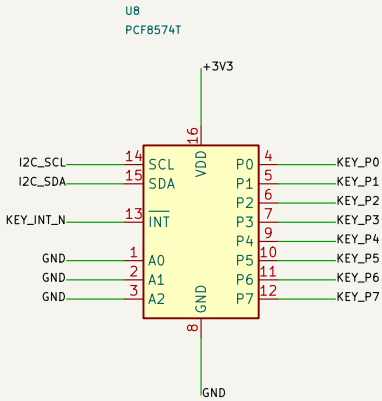
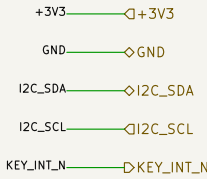
Reset isolation to 3.0 V logic



Sequence: logic stable -> reset pulse >=3us -> enable boost -> AFh.
Shutdown: AEh -> disable boost; keep logic powered.
Q3 removes the boost diode feed-through path when OFF.
Do not actively short OLED VCC/VDD to ground (SSD1309 guidance).
12.5 V nominal; retained 1.8M/200k and 910k IREF starting values.
Final glass current, contrast and boost load require validation.

GPIO10 HIGH asserts OLED reset; LOW releases it.
BS1 high; BS2 low; CS# low; SA0 low => I2C 0x3C.
D1 + D2 both connect to SDA (input + ACK output).
The SSD1309 is on the glass; DS1 is its 26-pad interface.
No RT9193-33 or TXB0108; shared I2C pull-ups are 3.0 V.
Exact panel/flex mechanics remain to be confirmed.

BREAKAWAY KEYBOARD / PCF8574T / 0x20



Place every component on this sheet on the detachable PCB section.
Only +3V3, GND, SDA, SCL and INT_N may cross the break line.
Keep connector pads/components clear of tabs, score and fracture area.
J4 (main board) <-> J5 (module): straight-through five-wire reconnect.
J3 is the requested 9-pin keyboard header and is DNP / user fitted.

100 kHz maximum. A2:A0 = 000, address 0x20 (PCF8574T, not 8574A).
P0..P7 are quasi-bidirectional; write 1 to use a pin as an input.
Header supports eight switch lines + common GND, or eight matrix lines.
Matrix mapping and debounce belong in firmware; pin 9 is ground.
R38/R39 optional pull-ups; leave DNP on the shared main-board bus.
Detached use: provide 3.3 V, ground and one suitable set of bus pull-ups.

Sheet: /Keyboard / breakaway / 0x20/
File: keyboard.kicad_sch

Title: Detachable PCF8574T keyboard interface

Size: A3
Date: 2026-09-20

Rev: B-draft
Id: 5/5